ChestVision: AI Chest X-Ray Disease Detection

A deep learning system that looks at a chest X-ray and predicts the probability of five conditions at once, with a heat map showing which part of the image drove each prediction.

What it detects

Atelectasis, Cardiomegaly, Consolidation, Edema and Pleural Effusion. This is a **multi-label** task, so one X-ray can have more than one condition.

How it works

1. An **EfficientNetB0** CNN is trained on the **CheXpert** dataset for multi-label detection.
2. A **FastAPI** backend loads the model and returns a probability for each condition.
3. A **Streamlit** frontend lets you upload an X-ray and see the predictions in real time.
4. **Grad-CAM** highlights the regions of the image the model used, so the result can be inspected and not just trusted.

Tech stack

Python, TensorFlow, Keras, EfficientNetB0, OpenCV, FastAPI, Streamlit

Results

- AUC / accuracy per condition: **[add your numbers]**
- Dataset size used for training and testing: **[add your numbers]**

Setup

```bash
git clone https://github.com/shreyadoiphode-09/[repo-name].git
cd [repo-name]
pip install -r requirements.txt
```

Start the backend and the app (replace with your own file names):

```bash
uvicorn [backend_file]:app --reload
streamlit run [frontend_file].py
```

Dataset

The CheXpert dataset is not included because of its size and licence. Download it from the official CheXpert page and place it in the folder your code expects.

Notes

This project is for learning and research. It is not a medical device and must not be used for diagnosis.

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